Shimura Varieties:
The Geometry Behind the L-Functions

Diego Rincón
Independent

Abstract

A Shimura variety is a locally symmetric space $\mathrm{Sh}(G, X)$ attached to a reductive group G over $\mathbb{Q}$ and a $G(\mathbb{R})$-conjugacy class X of Hodge structures. As a complex manifold it is $G(\mathbb{Q})\backslash X \times G(\mathbb{A}_f)/K$; as an algebraic variety it has a canonical model over the reflex field $E(G, X)$ by Shimura–Deligne. Shimura varieties are the geometric source of the L-functions throughout this body of work: the modular curve gives L-functions of elliptic curves (Wiles, BCDT), the Kuga–Sato variety gives Kato's Euler system [2], the unitary Shimura varieties give the periods of the arithmetic GGP conjecture [1], and the Siegel varieties give L-functions of GSp_{2n} used in the Skinner–Urban proof [2]. The Eichler–Shimura relation identifies Hecke eigenvalues with Frobenius eigenvalues on étale cohomology, realizing the arithmetic Langlands correspondence [4] in geometric form. We build the full structure: Shimura data, the variety, canonical models, Hecke correspondences, the Eichler–Shimura relation, and the position of Shimura varieties in the proofs of BSD, Iwasawa, and the Tate conjecture.

1 Shimura Data

Definition 1 (Deligne torus)**.** The *Deligne torus* is $\mathbb{S} = \mathrm{Res}_{\mathbb{C}/\mathbb{R}}\mathbb{G}_m$, the Weil restriction of scalars of $\mathbb{G}_m$ from $\mathbb{C}$ to $\mathbb{R}$. As an $\mathbb{R}$-group, $\mathbb{S}(\mathbb{R}) = \mathbb{C}^\times$. A *Hodge structure* on a real vector space V is a homomorphism $h : \mathbb{S} \to \mathrm{GL}(V_\mathbb{R})$; the associated Hodge decomposition is $V_\mathbb{C} = \bigoplus_{p,q} V^{p,q}$ where $\mathbb{C}^\times$ acts on $V^{p,q}$ by $z \mapsto z^{-p}\bar{z}^{-q}$.

Definition 2 (Shimura datum)**.** A *Shimura datum* is a pair (G, X) where G is a connected reductive algebraic group over $\mathbb{Q}$ and X is a $G(\mathbb{R})$-conjugacy class of homomorphisms $h : \mathbb{S} \to G_\mathbb{R}$ satisfying Deligne's axioms:

(SV1) The adjoint representation of G on $\mathrm{Lie}(G)_\mathbb{C}$ is of type $\{(-1, 1), (0, 0), (1, -1)\}$ (the Hodge decomposition is of the correct type for a Hermitian symmetric space).

(SV2) The element $\mathrm{ad}(h(i))$ is a Cartan involution of $G^{\mathrm{ad}}_\mathbb{R}$ (the associated symmetric space is a bounded domain).

(SV3) G^{ad} has no $\mathbb{Q}$-factor on which the projection of h is trivial (no compact factors).

The *Shimura variety* is the double coset space

$$\mathrm{Sh}_K(G,X)(\mathbb{C}) = G(\mathbb{Q})\backslash X \times G(\mathbb{A}_f)/K,$$

for a compact open subgroup $K \subset G(\mathbb{A}_f)$.

Example 3 (Modular curves)**.** Take $G = \mathrm{GL}_2$, $X = \mathbb{C} \setminus \mathbb{R}$ (the union of the upper and lower half-planes $\mathfrak{h}^{\pm}$). The homomorphism $h_\tau : \mathbb{S} \to \mathrm{GL}_{2,\mathbb{R}}$ corresponding to $\tau \in \mathfrak{h}$ maps $a + bi \mapsto \left(\begin{smallmatrix} a & b \\ -b & a \end{smallmatrix}\right)$. For $K = K(N) = \ker(\mathrm{GL}_2(\hat{\mathbb{Z}}) \to \mathrm{GL}_2(\mathbb{Z}/N))$:

$$\mathrm{Sh}_{K(N)}(\mathrm{GL}_2, \mathfrak{h}^{\pm})(\mathbb{C}) = \mathrm{GL}_2(\mathbb{Q})^+\backslash\mathfrak{h} \times \mathrm{GL}_2(\mathbb{A}_f)/K(N) \cong Y(N)(\mathbb{C}),$$

the modular curve of level N. Its compactification is $X(N)$.

Example 4 (Unitary Shimura varieties)**.** For integers $r, s \geq 0$, take $G = \mathrm{GU}(r,s)$ (the unitary similitude group of a Hermitian form of signature (r,s) over an imaginary quadratic field $K/\mathbb{Q}$). The associated Shimura variety parametrizes abelian varieties with (K,Φ)-multiplication of signature (r,s) (CM abelian varieties with prescribed Hodge type). These are the spaces appearing in the arithmetic GGP conjecture [1]: the periods $\int_{\mathrm{GU}(r,s)} \phi$ are the arithmetic data encoding the r-th derivative $L^{(r)}(E,1)$.

2 Canonical Models

Theorem 5 (Shimura–Deligne canonical models)**.** *For any Shimura datum (G,X) with reflex field $E = E(G,X) \subset \mathbb{C}$ (the field of definition of the Hodge cocharacter), the inverse system $\{\mathrm{Sh}_K(G,X)\}_K$ has a canonical model over E: a system of algebraic varieties over E whose complex points are $\mathrm{Sh}_K(G,X)(\mathbb{C})$, with $G(\mathbb{A}_f)$ acting on the system, and with the Galois action of $\mathrm{Gal}(\bar{E}/E)$ determined by the Shimura reciprocity law.*

Remark 6 (The reflex field). For modular curves, $E = \mathbb{Q}$. For $\mathrm{GU}(r,s)$ Shimura varieties associated to an imaginary quadratic field K, $E = K$. The canonical model is the reason Shimura varieties carry arithmetic: they are algebraic varieties over a number field, so Galois acts on their cohomology.

3 Hecke Correspondences and L-Functions

Definition 7 (Hecke operator)**.** For $g \in G(\mathbb{A}_f)$ and compact open K, the *Hecke operator* $T(g) : \mathrm{Sh}_K \to \mathrm{Sh}_K$ is the algebraic correspondence given by the double coset KgK: the diagram

$$\mathrm{Sh}_{K\cap gKg^{-1}} \xrightarrow{[g]} \mathrm{Sh}_K \quad \text{and} \quad \mathrm{Sh}_{K\cap gKg^{-1}} \xrightarrow{\mathrm{pr}} \mathrm{Sh}_K.$$

For $G = \mathrm{GL}_2$, $g = \left(\begin{smallmatrix} p & 0 \\ 0 & 1 \end{smallmatrix}\right)$, and $K = K_0(N)$, this is the classical Hecke operator T_p. The Hecke algebra $\mathbb{T} = \mathbb{Z}[T_p : p \nmid N]$ acts on $H^1_{\text{ét}}(\mathrm{Sh}_K \otimes \bar{\mathbb{Q}}, \mathbb{Z}_\ell)$.

Theorem 8 (Eichler–Shimura relation)**.** *For the modular curve $X_0(N)$ and a prime $p \nmid N\ell$, the Hecke operator T_p and the Frobenius $\mathrm{Frob}_p \in G_{\mathbb{Q}}$ satisfy, on $H^1_{\text{ét}}(X_0(N)_{\bar{\mathbb{Q}}}, \mathbb{Q}_\ell)$:*

$$T_p = \mathrm{Frob}_p + p \cdot \mathrm{Frob}_p^{-1}.$$

Remark 9 (Why this is the Langlands correspondence). The Eichler–Shimura relation identifies the Hecke eigenvalue $a_p(f)$ of a newform f with the trace of Frobenius on the ℓ-adic Galois representation ρ_f: $a_p(f) = \mathrm{tr}(\mathrm{Frob}_p \mid \rho_f)$. This is the arithmetic Langlands correspondence for GL_2 in geometric form. The automorphic data (Fourier coefficients of f) and the Galois data (Frobenius traces of ρ_f) are identified via the geometry of the modular curve. The Shimura variety is the bridge between the automorphic grammar and the Galois grammar [5].

4 The Kuga–Sato Variety and Kato's Euler System

Definition 10 (Kuga–Sato variety)**.** For the modular curve $X_0(N)$ with universal elliptic curve $\mathcal{E}$, the *Kuga–Sato variety* of weight r is the r-fold fiber product:

$$W_r = \mathcal{E} \times_{X_0(N)} \mathcal{E} \times_{X_0(N)} \cdots \times_{X_0(N)} \mathcal{E} \quad (r \text{ factors}).$$

W_r is a smooth projective variety of dimension $r + 1$ over $\mathbb{Q}$.

Theorem 11 (Deligne, Scholl)**.** *The motive $h^r(W_r)$ contains the motive $M(f)$ attached to a newform f of weight $r + 2$. The Galois representation on $H^r_{\acute{e}t}(W_{r,\bar{\mathbb{Q}}})$ contains ρ_f as a direct summand.*

Theorem 12 (Kato's Euler system via Kuga–Sato [2])**.** *Kato constructs his Euler system from algebraic K-theory elements in $K_2(W_1(\mu_{p^n}))$ (modular units on the Kuga–Sato variety at level p^n). The norm-compatibility follows from the geometry of the tower $\{W_1(\mu_{p^n})\}_n$ over the cyclotomic tower. The resulting Euler system gives the bound* $\mathrm{char}_\Lambda(X_\infty) \mid (L_p(E))$ *in the Iwasawa algebra.*

5 Shimura Varieties and BSD

Theorem 13 (Modularity: Wiles, Taylor–Wiles, BCDT)**.** *Every elliptic curve $E/\mathbb{Q}$ is modular: there exists a newform $f \in S_2(\Gamma_0(N))$ (for $N = N_E$ the conductor) such that $L(E, s) = L(f, s)$. Geometrically, there is a surjection*

$$\phi : X_0(N) \twoheadrightarrow E$$

of algebraic curves over $\mathbb{Q}$, the modular parametrization. The Heegner points [2] on E are the images under ϕ of CM points on $X_0(N)$.

Remark 14 (The chain from Shimura to BSD). The chain of implications is:

$$X_0(N) \xrightarrow{\phi} E \xrightarrow{\text{Heegner}} y_K \in E(K) \xrightarrow{\text{Gross-Zagier}} L'(E, 1) \neq 0 \xrightarrow{\text{Kolyvagin}} \mathrm{rank}\, E(\mathbb{Q}) = 1.$$

Every step goes through the Shimura variety $X_0(N)$. BSD (rank ≤ 1) is a consequence of the geometry of the modular curve.

6 Shimura Varieties and the Tate Conjecture

Theorem 15 (Tate for abelian varieties via Shimura [3])**.** *The Shimura variety for* $(G, X) = (\mathrm{GSp}_{2g}, \mathfrak{H}_g)$ *(the Siegel modular variety of genus* g*) parametrizes principally polarized abelian varieties of dimension* g*. The Tate conjecture for these abelian varieties (Faltings' theorem) is proved using the structure of the Shimura variety: the Hecke correspondences generate the endomorphism algebra, and the semisimplicity of the Galois representation on* $T_\ell(A)$ *follows from the Hodge-theoretic structure of the Shimura datum.*

Theorem 16 (Tate for K3 surfaces via Kuga–Satake [3])**.** *The Kuga–Satake construction associates to a K3 surface* X *an abelian variety* $A = A(X)$ *such that* $H^2_{\acute{e}t}(X) \hookrightarrow H^1_{\acute{e}t}(A) \otimes H^1_{\acute{e}t}(A)$*. The pair* $(A, H^2(X))$ *arises from a Shimura variety for a spin group. The Tate conjecture for K3 surfaces (André, Kim–Madapusi Pera) follows by reducing to the Tate conjecture for abelian varieties via this embedding.*

7 Position in the Body

Theorem 17 (Shimura varieties as the source)**.**

Result in body	*Shimura variety used*	*Role*
BSD rank ≤ 1 *[1, 2]*	*Modular curve* $X_0(N)$	*Source of Heegner points*
Kato Euler system [2]	*Kuga–Sato variety* W_r	*Source of* K*-theory elements*
Skinner–Urban [2]	*Siegel variety* $\mathrm{Sh}(\mathrm{GSp}_4)$	*Eisenstein series*
Arithmetic GGP [1]	*Unitary Shimura* $\mathrm{Sh}(\mathrm{GU}(r, s))$	*Period integrals*
Tate for K3 [3]	*Spin Shimura variety*	*Kuga–Satake embedding*
Langlands [4]	*All Shimura varieties*	*Geometric realization*

Shimura varieties are not one tool among many. They are the geometric common source: every L*-function, every Euler system, every period in this body of work lives on or comes from a Shimura variety. The Arithmetic Wall [7] is the failure to extend the geometry of Shimura varieties from local fields (perfectoid [6]) to the global field* $\mathbb{Q}$*.*

References

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